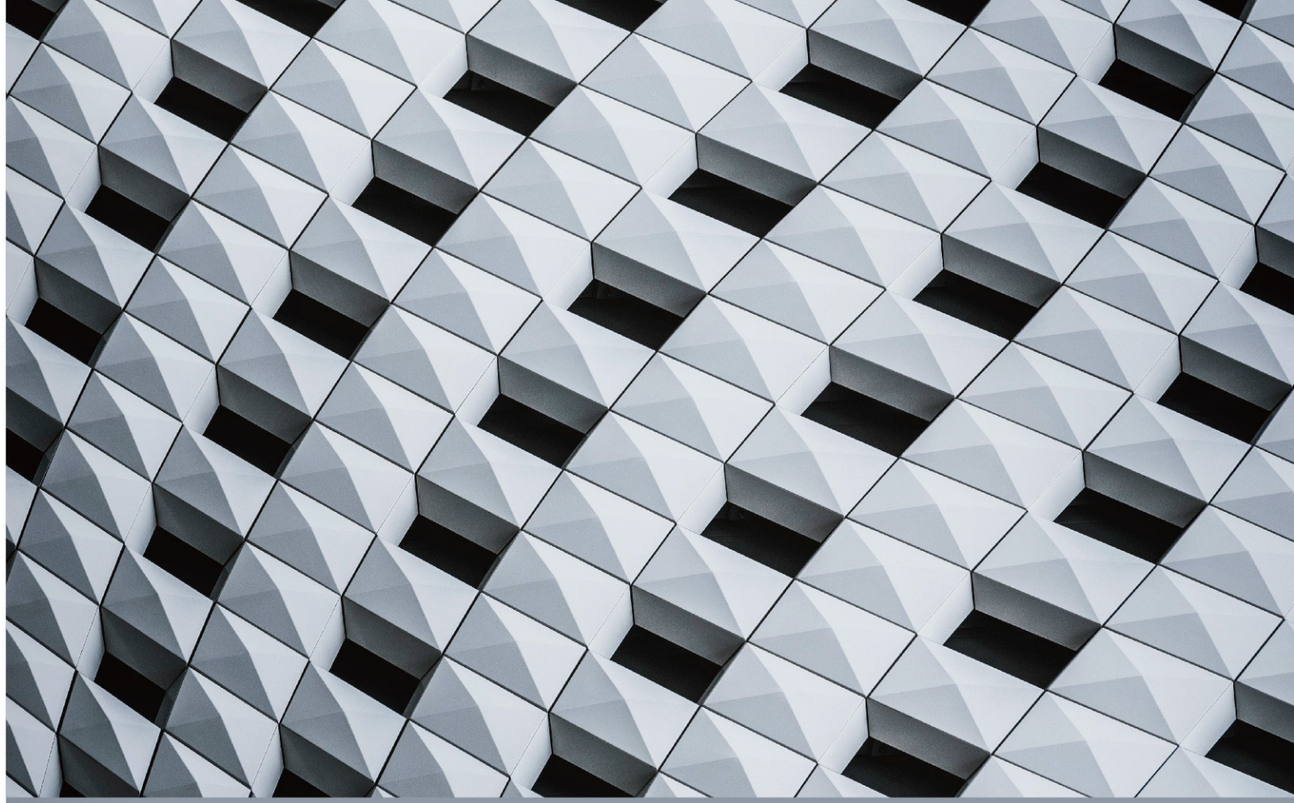
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TECHNICAL MANUAL

Feiteng Metal Composite Panel

FEITENG

FEITENG

FEITENG

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FEITENG aluminium composite panels are clearly divided into two product series based on different core material characteristics, serving the two major fields of building curtain walls and signage/advertising. The core material determines the panel's mechanical properties, fire rating, and processing methods. Users should select the appropriate series according to the specific application scenario and strictly follow the processing guidelines provided in this manual.

Product Series and Core Material Characteristics

DIPPER Series

Dedicated to Building Curtain Walls

- Core material type: A2 grade non-combustible core or FR flame-retardant core
- Application scope: building curtain walls, exterior cladding, public buildings with strict fire protection requirements
- Processing characteristics: The core material has high hardness, so cemented carbide tools must be used for cutting, grooving, and bending. V-grooving before bending is recommended, with an optimal V-groove angle of 95–110°. A polyethylene core thickness of 0.2–0.4 mm (excluding the aluminum layer) should be retained at the groove bottom, which effectively prevents bending cracks caused by damage to the aluminum skins. Sharp molds should be selected for drilling and punching to reduce core powder or debris.



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LYRA Series

Dedicated to Signage and Digital Printing

- Core material type: Conventional low-density polyethylene (LDPE) core
- Application scope: digital printing, signage, display boards, interior decoration
- Processing characteristics: The core material is relatively soft, allowing easy and fast cutting, but it is prone to melting or burr formation. For saw cutting, carbide-tipped blades designed for aluminum/plastic are recommended. For punching or perforating, the hole diameter should not be less than 4 mm, and the web distance between holes should not be less than 4 mm. Because the core is temperature-sensitive, it is recommended that the processing environment temperature be no lower than 16°C to prevent brittle cracking during bending at low temperatures.



General Processing Recommendations

Regardless of which series is selected, FEITENG aluminium composite panels offer good machinability. However, the following common principles should be observed:



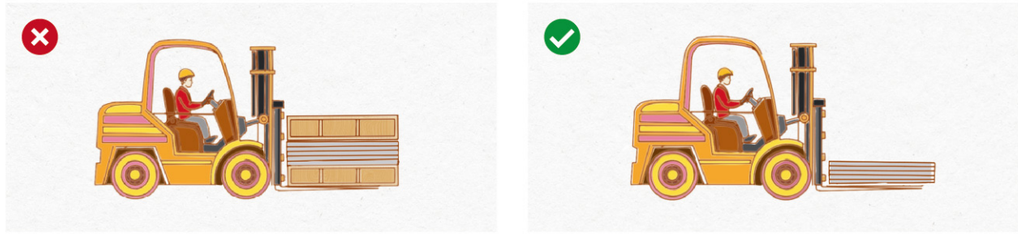
- Cutting: CNC saw cutting (vertical panel saw, circular saw) can be used. Laser cutting should be avoided (the fumes may damage optical components). Shearing may cause minor edge deformation and is not suitable for exposed cut edges.
- Grooving and bending: Only applicable to the architectural grade (DIPPER) and advertising grade (LYRA) series with aluminum thickness above 0.3mm. Grooving must keep the back aluminum skin intact. We recommend retaining at least 0.2–0.4mm of core thickness (excluding the aluminum layer).
- Thermal expansion: FEITENG aluminium composite panels expand approximately 2.4 mm per linear metre for a 100°C temperature change and that fabrication and fixing methods should allow for this movement. We recommend 4.0 mm per linear metre for a 100°C temperature change.
- Cleaning: Remove cutting chips promptly after processing to prevent dents or surface marks.
- Safety: Appropriate protective equipment should be worn for all processing operations, and the equipment instructions should be followed.

This manual aims to provide clear, actionable technical guidance for fabricators, installers, and designers. For special processing requirements, please contact the FEITENG technical support team.



Protection and Stacking Standards

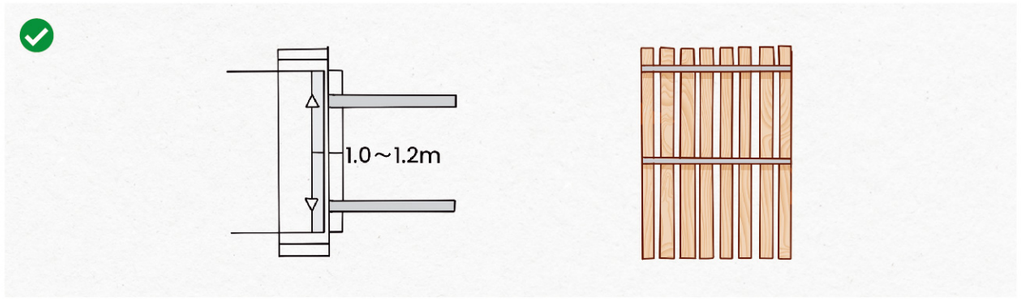
Stacking is a critical step in the transportation, storage, and installation of Feiteng metal composite panels, directly affecting the panel's surface quality, structural stability, and subsequent construction results. Improper stacking methods can easily lead to issues such as scratches, deformation, delamination, and aging. Feiteng has established strict and scientific stacking specifications and standards. For more details, please visit www.feiten-gacp.com to refer to the Feiteng Metal Composite Panel Protection and Stacking Standards.



Packed panels should be transported horizontally. Do not place other heavy objects or different materials.

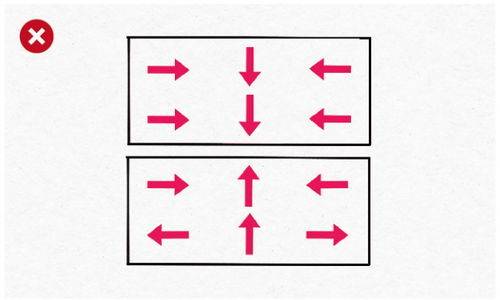


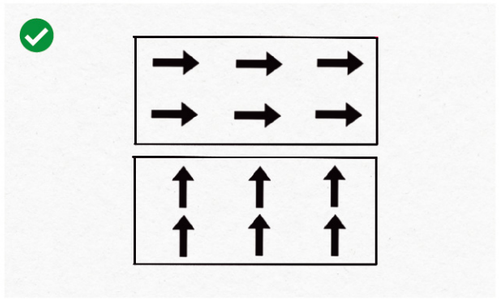
During transportation, the outer packaging should be clearly marked with labels such as "Handle with Care," "Keep Dry," "Do Not Use Hooks," and "This Side Up."



When the distance between the forklift forks is fixed at 1.0–1.2 m, pallets with a fork hole center distance of the same range (1.0–1.2 m) should be selected or customized. Handle pallets carefully during transportation. If any pallet is damaged by moisture, it must be reported immediately and confirmed by the carrier.

Processing and Installation





Before processing, clean the workbench, temporary supports, and both sides of the FEITENG panel. Pay particular attention to cutting chips from processes such as sawing, milling, and drilling, as well as any debris particles trapped between the panel and tools, as these may cause surface damage.

- To avoid differences in reflection (for finishes like metallic or brushed surfaces), it is recommended to install panels in the same orientation as indicated by the directional marks on the protective film. For optimal color uniformity, it is strongly recommended to use panels from the same production batch for a single project.
- Remove the protective film as soon as possible after installation. If the protective film is exposed to outdoor conditions for an extended period, it may become extremely difficult to remove later.

FEITENG

TECHNICAL DATA

DIPPER Series (FR CORE & A2 CORE PANEL)

Architectural Facade Panels | Curtain Wall Applications

		Alu. Thickness: 0.5mm	
Panel thickness tolerance	**Units**	**4mm**	
Alu thickness tolerance	mm	±0.02	
Alu grade		3003, 5005	
Composite Panel Size			
Panel weight	kg/m²	≥7.5	≥8.7
Panel length tolerance	mm	±3	
Panel width tolerance	mm	±2	
Panel thickness tolerance	mm	±0.2	
Mechanical Property			
Bending strength	MPa	≥100	
Penetration resistance	KN	≥7	
Shear strength	MPa	≥22	
Tensile strength of Alu	MPa	R_m≥130	
0.2% yield stress	MPa	$R_{P0.2}$≥90	
The elongation of Alu	MPa	A_{50}≥4	
Liner thermal expansion	%	≤0.0004	
Resistance to temperature difference	℃-1	≥100	
Flexural elastic modulus	℃	≥2.0*10⁴	
Peeling force	N/mm	Average≥110, Minimum≥100	
Alloy state	/	H24	
Composite Panel Core		FR	A2
Core density	g/cm³	1.55~1.60	1.95~2.00
Fire Rating Classification		EN13501-1/Class B-s1,d0	EN13501-1/Class A2-s1,d0
Coating		PVDF	
Coating thickness	μm	Single avg≥16 min≥14; Double avg≥25 min≥23; Third avg≥32 min≥30	
Glossiness	%	≥25	
Coating hardness	/	≥HB	
Acoustic Performance			
Sound absorption factor	α	0.05	
Sound insulation factor	dB	26	
Thermal Performance			
Heat transfer coefficient	W/mK	5.53	5.8

LYRA Series

Digital Printing | Signage | Display | Interior Applications

	Units	LITE Alu. Thickness: 0.2mm				PLUS Alu. Thickness: 0.25mm				PREMIUM Alu. Thickness: 0.3mm			
Panel thickness tolerance	mm	2	3	4	6	2	3	4	6	2	3	4	6
Alu thickness tolerance	mm	±0.02				±0.02				±0.02			
Alu grade		1100				1100				1100, 3003, 5005			
Composite Panel Size													
Panel weight	kg/m²	≥2.93	≥4.08	≥5.23	≥7.53	≥3.09	≥4.24	≥5.39	≥7.69	≥3.25	≥4.40	≥5.55	≥7.85
Panel length tolerance	mm	±3				±3				±3			
Panel width tolerance	mm	±2				±2				±2			
Panel thickness tolerance	mm	±0.2				±0.2				±0.2			
Mechanical Property													
Bending strength	MPa	/	≥50			/	≥50			/	≥50		
Penetration resistance	KN	/	≥3			/	≥3			/	≥5		
Shear strength	MPa	/	≥12			/	≥12			/	≥12		
Tensile strength of Alu	MPa	≥145											
Alu Proof stress(0.2%)	MPa	≥115											
The elongation of Alu	%	≥2											
Liner thermal expansion	mm/m	≤2.4(100°C) Aluminum											
	mm/m/°C⁻¹	≤0.0004 Panel											
Thermal deformation temperature	°C	≥85											
180° peel strength	N/mm	≥4				≥4				≥5			
Heat resistance water	/	No change											
Composite Panel Core													
Core density	g/cm³	1.15~1.20											
Fire Rating Classification		EN13501-1/Class C-s1,d0											
Coating		Polyester											
Coating thickness	μm	Single avg≥16 min≥14; Double avg≥25 min≥23; Third avg≥32 min≥30											
Glossiness	%	≥25											
Coating hardness	/	≥HB											
Acoustic Performance													
Sound absorption factor	α	0.05											
Sound insulation factor	dB	23	25	26	27	23	25	26	27	23	25	26	27
Thermal Performance													
Heat transition coefficient	W/mK	5.98	5.65	5.54	5.34	5.98	5.65	5.54	5.34	5.98	5.65	5.54	5.34

FEITENG

Processing Method

FEITENG metal composite panels can be easily processed using standard metalworking tools for cutting, punching, grooving and bending. Based on the desired architectural façade design, they can achieve various shapes, including diverse curved surface designs.

Tools

CUTTING & PUNCHING



table saw



punching



square shear



water jet cutting machine



hand router



jig saw



turret puncher

BENDING



press brake



3-roll bender



folding machine

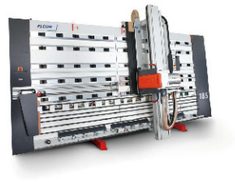
ROUTING & FOLDING



grooving machine



hand router



panel saw



CNC router

Processing Method

CUTTING & PUNCHING

FEITENG

Saw Cutting

FEITENG panels can be easily cut and shaped using a variety of sawing equipment, including vertical panel saws, circular saws, and jigsaws. For optimal cutting performance and clean, burr-free edges, it is recommended to use carbide-tipped saw blades specifically designed for aluminum or plastic materials. These blades minimize melting, chipping, and blade binding, ensuring smooth, precise cuts even on complex contours or large-format panels.

Example of suitable saw blade

Blade diameter	225m
Number of teeth	80 to 100
Cut width	2.0 to 2.6mm
Rake angle	10°
Tip	Carbide

Operating Conditions

Rotation of saw blade	2000-4000 rpm
Feed speed	10-30 m/min



Notes on Saw Cutting

- Always perform cutting operations with the effective side (finished surface) facing upward. This helps prevent scratches on the panel surface and avoids peeling off the protective film.
- After cutting, carefully remove cutting chips and debris from the FEITENG Aluminum Composite Panel. Residual chips may cause dents or surface marks during subsequent storage or assembly.
- When the saw blade becomes dull, sharpen or replace it immediately. A dull blade will produce large burrs or cause distortion along the cut edge, compromising both appearance and dimensional accuracy.

Cutting

Shear Cutting

FEITENG panels can also be cut using a conventional guillotine (power shear). To achieve the best possible cut quality, the following prerequisites must be met:

- A shearing angle of approximately 1.5°
- Minimum blade clearance – verified by a successful paper test (i.e., the sheared paper edge shows a clean cut without excessive.)
- To prevent damage to the cover sheet (top decorative layer), it is advisable to fit the guillotine's hold-downs (or pressure pads) with protective rubber pads.



⚠ Important Note

- Cutting or shearing FEITENG panels by these methods is not suitable for decorative applications where the cut edge will be fully visible, such as in riveted facades. For such visible-edge applications, alternative edge-finishing methods (e.g., routing, edge banding, or factory-prefinished edges) should be considered.
- Mechanical shearing using guillotines may cause minor deformation of the aluminium skin at the cut edge. Where exposed edges are required, saw cutting or CNC routing is recommended to ensure a clean edge finish.

Trimming

In saw cutting, burrs typically appear on both sides of the cut edge. In shear cutting, either a droop (rounded edge) or a burr may be present on each edge, depending on blade clearance.

If panels are to be installed with the cut edge exposed, special attention must be paid to the edge condition.

- For saw cutting, keep the saw blade sharp to achieve a clean, sharp cut.
- For shear cutting, adjust the die clearance properly to minimise deformation.

Generally, edge condition is more critical for interior applications than for exterior ones. In some cases, trimming of the edge is necessary after cutting. Trimming can be performed using a trimmer, a plane, or sandpaper.

⚠ For Solid and Metallic Colors

- Deep trimming, such as chamfering, can enhance visual appearance. Use a trimmer fitted with a ball-bearing chamfering bit, or a plane designed for woodworking. When using a plane, a guide ruler helps ensure a uniform edge finish.

⚠ For Stone and Timber Finishes

- Deep trimming is not suitable, as it may damage the aesthetic integrity of the stone or timber surface. If cut edges in stone or timber finishes are sharp enough to hurt a finger, lightly sand them with fine sandpaper to dull the edge. Normally, the droop edge produced by shear cutting is sufficiently mild to ensure safe handling.

Curved Cutting

A hand router or trimmer can be used to cut FEITENG panels along curved lines. A guide template helps to stabilize the tool during this operation. A jigsaw is also useful for cutting complex shapes.





⚠ Notes on Using a Guide Plate (Template)

- Place an appropriate guide plate (template) on the effective side (finished surface) of the FEITENG panel, then perform the routing operation by guiding the router/trimmer along the inside edge of the template.
- Any particles or debris trapped between the template and the effective surface may cause dents or scratches on the panel. Keep both surfaces clean before and during the operation.

Water Jet Cutting

During water jet cutting, the plunge cut (piercing at the starting point) may cause a certain degree of delamination between the aluminium skin and the core material. Therefore, the plunge should be made either in a disposable (sacrificial) area or starting from the panel edge. Once the water jet has fully penetrated the panel, it can then cut the FEITENG panel without further delamination issues.



Laser Cutting

Based on our tests, we have concluded that FEITENG panels are not suitable for laser cutting, as the fumes generated during the process may damage the sophisticated optical components of laser systems.

Cutting

Punching

FEITENG panels of any thickness can be punched using conventional sheet metal punching machines.

For clean cuts, use sharp tools and dies with minimal cutting clearance (e.g., 0.1 mm).

This punching process may cause slight deflection (bowing) of the cover sheet around the punched hole, which is normal and typically does not affect panel performance.



⚠ Additional Notes on Punching

Punching Edge Distance Recommendation

- The distance from the punched hole to the panel edge should not be less than the panel thickness or 15 mm (whichever is larger), to avoid edge cracking or deformation.

Die Clearance

- Excessive die clearance will produce burrs, while insufficient clearance may aggravate cover sheet deflection. It is recommended to perform a trial punch first to confirm hole edge quality.

Preventing Surface Layer Damage

- During punching, the decorative side (finished surface) of the panel should face the punch direction. Additionally, protective rubber or felt pads should be placed on the hold-downs (pressure pads) to reduce pressure marks.

Batch Processing Considerations

- Punching tools should be replaced promptly when worn. Dull tools will lead to increased burrs on the hole edge, surface layer peeling, or even tearing of the aluminium skin.

Perforating (for interior applications only)

FEITENG panels can be perforated using CNC punching machines. This process is commonly used for interior and ceiling design applications.

Technical recommendations

- Minimum hole diameter: 4 mm
- Minimum web (distance between adjacent hole edges): 4 mm

For best results, use a single-punch die for precise holes; multi-station machines are more economical for high-volume production.

After perforation

- The flatness of the panel may be affected. If required, further processing (e.g., roller leveling or vacuum flattening) may be necessary.
- To protect the decorative surface, place protective rubber or felt pads on the machine's hold-downs.
- When perforating near the panel edge, maintain a minimum distance of 15 mm from the edge to avoid cracking or deformation.

FEITENG

Turret Puncher (CNC Turret Punch Press)

A CNC turret punch press (computer-controlled) can also be used for perforating FEITENG panels. The recommended clearance between punch and die is 0.1 mm or smaller (approximately 2% of the total panel thickness). A slight droop (rounded edge) may appear on the punched hole edge, which is normal and does not affect interior applications.

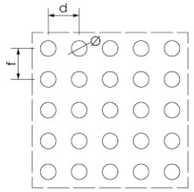


CNC turret punch press is suitable for high-volume, multi-shape, and high-efficiency perforation processing. Its advantages include:

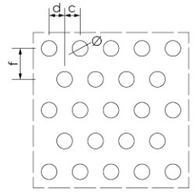
- Automatic tool changing allows continuous operation without manual intervention.
- Can punch round, square, hexagonal, elongated, and decorative holes in one setup.
- Ideal for high-volume production due to fast cycle times.
- Minimal burrs and consistent edge finish with proper clearance (0.1 mm or smaller).
- Easily programs straight or staggered hole arrangements.

Punching Parameters

Classification	Parameter	Notes
Hole diameter (a)	≥4.5mm	
Hole spacing (d, f, e)	≥10mm	
Transmittance	≤35%	Transmittance=(punching area/total area) x 100%



Distribute Even



Distribute Cross

Processing Method

BENDING

FEITENG

Bending with press brake

FEITENG panels are similar to metal sheets and can be easily formed using a bending machine (press brake). When forming with a bending machine, grooving is usually performed first, followed by the air bending process.



The minimum bendable radius with press brake is as follows:

Bending Direction	Minimum bendable radius
Traverse	50mm
Parallel	80mm

⚠ Detailed recommendations

- V-die opening: V-groove bottom width: We recommend a V-groove bottom width of 2.0–4.0 mm. The wider the groove bottom, the lower the risk of cracking, but the bending radius (R corner) will increase accordingly.
- Minimum bend radius (cold bending, no routing): ≥ 350 mm. Below this value, visible wrinkles may appear on the aluminium surface.
- Springback compensation: Aluminium composite panels exhibit springback. Overbend by 1–2 degrees to achieve the desired angle.

⚠ **Notes on press brake bending** (original points refined)

- "Traverse" and "Parallel" on the protective film show the bending direction relative to the rolling direction.
- The minimum bendable radius is the limit at which visible wrinkles appear on the aluminium surface.
- Use a top die with a radius close to the desired bend radius. If the top die radius is too small, local bending may fall below the limit, causing wrinkles.
- Always use a urethane pad or rubber mat between the panel and the bottom die.
- Ensure the top die is scratch-free; polish and wipe it. Perform bending without peeling off the protective film.

Bending

Bending with a 3-roll bender

Manual or electric-drive 3-roll benders can be used to curve FEITENG panels (cylindrical bending). The minimum bendable radius is normally 350 mm (radius), but this depends on the bender length and machine type. (See the example relationship chart in the original manual.)



Roll length (mm)	Minimum radius (mm R)
500	120
1000	150
2000	180
2500	200

⚠ Detailed recommendations

- Roll surface preparation: Wipe the rolls carefully before bending to remove any dirt or particles that could dent the panel.
- Edge preparation: Remove any burrs from the panel edges – they can cause dents during rolling.
- Cleanliness: Remove cutting chips stuck on the panel and flatten any wrinkles in the protective film; both can create surface marks.
- Roll clearance adjustment: Do not tighten the panel tightly between the rolls. If the machine has fixed roll clearance, adjust it to panel thickness + approx. 0.5 mm.
- Notch sequence: If a notch is required in a curved panel, make the notch after bending (or make edge bends after curving). Making the notch before bending will distort the curve.
- Small radius bending: For radii approaching the minimum limit, perform gradual bending by slowly adjusting the elevation of the bending roll (multiple passes).
- Straight edge reduction: A subsidiary (sacrificial) sheet can reduce the straight portion near the leading/trailing edge, but some straight section always remains. If a continuous curve is needed near the edge, perform additional edge bending after the main curving operation.

FEITENG

Bending with a folding machine

In folding machines, the panel is clamped between two cheeks (upper and lower clamping beams). The projecting edge is bent around the upper clamping cheek (or a former) using a movable swivel bar. The bending radius is determined by interchangeable formers attached to the upper clamping cheek.



⚠ Detailed recommendations

- Clamping pressure: Adjust clamping force to hold the panel firmly without crushing the core or leaving jaw marks. Use rubber-lined clamping cheeks when possible.
- Bending sequence: For multiple bends (e.g., a tray shape), perform bends in order from the centre outward.
- Minimum radius: Feiteng composite panels must be V-grooved (milled) before bending, leaving 0.2–0.4 mm of polyethylene core material (excluding the aluminum layers) at the groove bottom, with a groove bottom width of 2.0–4.0 mm, and a minimum bending radius of ≥2mm.
- Protective film: Keep the protective film on the finished side during folding to avoid scratches from the clamping cheeks.

Minimum Bending Radius Clarification

This section clarifies the confusion between cold curving (without routing) and routed folding.

⚠ Routed folding (V-grooving)

- Before bending, Feiteng composite panels must be V-grooved (routed) on the back side. The groove bottom shall retain 0.2–0.4 mm of polyethylene core material (excluding the aluminium skins), with a groove bottom width of 2.0–4.0 mm, ensuring a minimum bending radius ≥ 2 mm.
- After grooving, the panel can be folded at any angle. We recommend a bending radius R ≥ 2 mm; a smaller radius may easily cause material cracking.

Processing Method

ROUTING & FOLDING

FEITENG

CNC Router

A CNC router can be used for cutting and grooving FEITENG panels. Because the entire process is controlled by a computer program, the CNC router is especially suitable for repeating the same processing tasks with high precision and consistency. The appropriate router bit types and operating conditions are the same as those for hand routers. We particularly recommend using a CNC router for grooving operations.



Handy grooving machine

A hand grooving machine can be used to groove FEITENG panels. Always secure the panel firmly and use a guide rail or straightedge for straight grooves. Keep the protective film on the finished side to prevent scratches.





Cutter balde

Outside diameter	110-120mm
Number of teeth	4~8
Material	Carbide tip

Operating Conditions

Rotation	5,000-9,000 rpm
Feeding speed	5-20 m/min

Handy router

Milling cutter type: Carbide V-grooving cutter (common angles include 90°, 135°, 150°, etc.) or radius cutter. Determine the design angle of the finished bent product and select a V-cutter that matches the final bending angle of the processed panel.

When grooving straight lines, use a guide rail or straightedge.



Router bit

Number of teeth	2-4
Material	Carbide tip

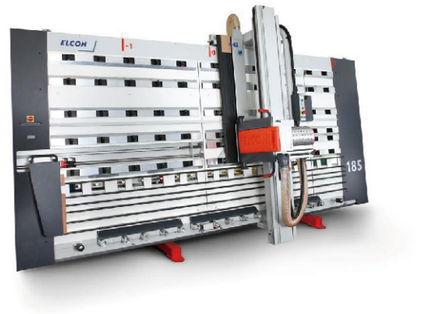
Operating Conditions

Rotation	20,000-30,000 rpm
Feeding speed	3-5 m/min

Routing & Folding

Panel saw

Efficient grooving work is possible with a panel saw.
Typical conditions are as follows:



Cutter balde

Outside diameter	220 mm
Material	Carbide tip

Operating Condition

Rotation	2
Feeding speed	30 m/min

Key Technical Points

FEITENG panels can be routed and folded using a flatbed CNC routing machine, a vertical panel sawing machine, or a handheld "V-grooving" router. The panels can then be bent and formed into cassette trays and integrated into exterior façade or curtain wall systems.

A "V-shaped" or "rectangular" groove shall be machined on the back side of the panel along the intended bending line. During grooving, the groove must not cut completely through the panel. A core thickness of 0.2–0.4 mm must be retained at the groove bottom to provide sufficient material for bending. This remaining thickness helps prevent cracking or crazing of the paint layer and aluminum skins during the bending process.

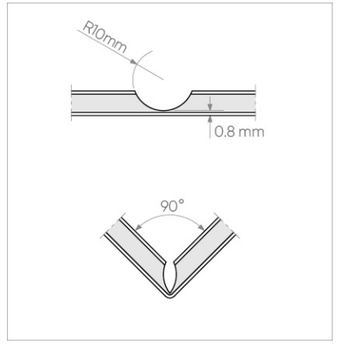
⚠ Additional recommendations

Minimum remaining thickness

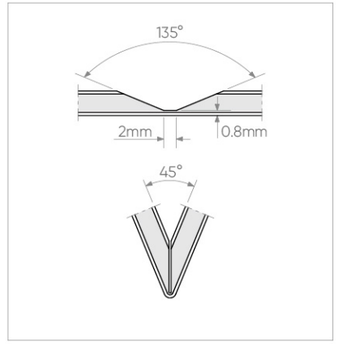
- For panels with an aluminum skin thickness of 0.3 mm, the total remaining thickness at the groove bottom (back aluminum skin + core) should be approximately 0.5–0.8 mm. It is recommended to retain at least 0.2–0.4 mm of core thickness (excluding the aluminum layer). This ensures both bending strength and sufficient flexibility.

Groove angle

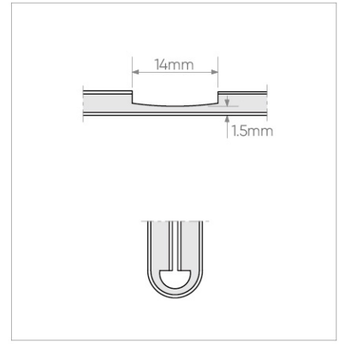
- For 90° folding (sharp corner), a V-groove angle of 90° is typically used.
- For folding angles other than 90°, the groove angle should match the desired bending angle (e.g., 60° groove for 60° folding).
- Rectangular (straight) grooves are used when a larger bending radius is acceptable or when the panel will be cold-bent without folding to a sharp corner.



Spherical milling slot for 90° bends.



V-shaped milling slot for 135° bends.



Arc milling slot for full 180° bends (not possible on FEITENG A2 panel).

Bending speed

- When folding a routed panel, perform the bending smoothly and in one continuous motion. Avoid sudden force or jerky movements. Recommended speed: slow and steady (e.g., complete the bend within 2–3 seconds for a typical cassette edge). Do not repeat the bending motion (back and forth), as this will fatigue the remaining aluminium skin and cause cracking.

Temperature consideration

- For best results, ensure the panel temperature is above 16℃ before routing and folding.

Tooling

- Use a sharp carbide-tipped routing bit. Dull bits will create rough groove surfaces and increase the risk of delamination or uneven folding.

JOINING METHOD

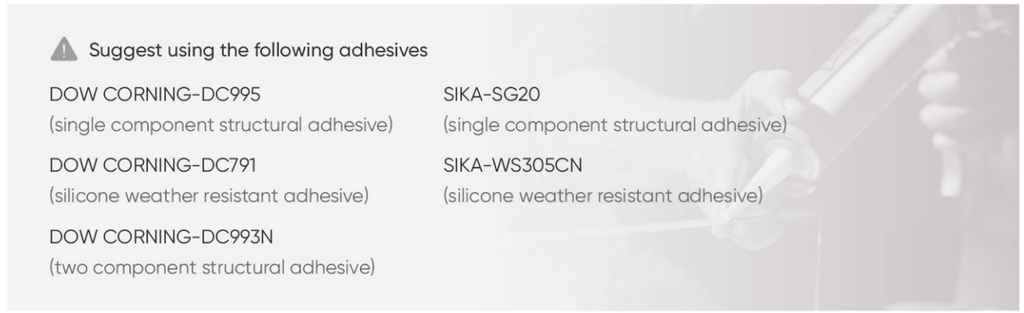
FEITENG

Riveting

The connection of FEITENG metal composite panels is done using rivets. Commonly used rivet specifications include M411, M511, although other models can be chosen based on requirements. Rivet connections require holes to be drilled in the composite panel. The distance from the center of the hole to the edge of the panel should not be less than twice the panel thickness. A gap of no more than 0.3mm should be left between the hole and the rivet to avoid compressing and deforming the composite panel.

Adhesion

When the size of the FEITENG metal composite panel is relatively large, structural reinforcement should be applied to the back of the panel based on the design wind load value of the project. The reinforcement material should be aluminum profiles (which can be U-shaped, C-shaped or square tubes). The aluminum profiles and the back of the panel should be effectively bonded using a two-component silicone structural adhesive (and, if necessary, combined with 3M double-sided adhesive).



Suggest using the following adhesives

DOW CORNING-DC995
(single component structural adhesive)

DOW CORNING-DC791
(silicone weather resistant adhesive)

DOW CORNING-DC993N
(two component structural adhesive)

SIKA-SG20
(single component structural adhesive)

SIKA-WS305CN
(silicone weather resistant adhesive)

Notes on Adhesives (Revised)

- Prior to adhesion work, remove all foreign matters such as dust, particles, grease, water, etc. from the area to be adhered.
- Select the most appropriate adhesive that ensures the necessary adhesion strength under the intended atmospheric conditions. The adhesion power depends on the surface condition of the substrate.
- Always follow the adhesive manufacturer's recommended curing conditions and surface preparation requirements.
- When FEITENG panels are adhered to a different material, the panel may show deflection due to differences in thermal expansion or dimensional change of the material. Pre-test the adhesive before fabrication and installation.
- Some adhesives may cause distortion after hardening due to shrinkage, as shown in the diagram. Therefore, pre-testing is necessary for certain types of adhesives. Generally, some epoxy adhesives, polyurethane adhesives, and silicone adhesives may show this kind of distortion. This distortion is usually very slight and sometimes not visible on low-gloss or matte finishes.

Joining Method

Core Welding

One end of a FEITENG panel can be adhered to another end of a FEITENG panel by welding the core using hot melt adhesive (glue). Prior to heating the glue stick, the core surface must be preheated to ensure good adhesion. Normally, mechanical reinforcement is necessary after welding.

Double-sided Tape

Double-sided tape, such as 3M™ VHB™ tape, is effective for joining FEITENG panels to other materials. VHB tape simplifies the joining process, and the adhesive nature of such tapes allows a certain degree of movement between the two adhered materials.

Hook & Loop Fastener

Hook-and-loop fasteners (e.g., Velcro® tape) are useful for guide signs and displays. This type of fastener is removable and reusable.

Sealant

To ensure waterproofing of joints between panels, a sealing material is normally used. The sealant must meet the performance requirements of the intended environmental conditions. Silicone, modified silicone, polysulfide, and polyurethane sealants are commonly used.

FEITENG

SURFACE TREATMENT



Surface Treatment

Digital Printing

The polyester-lacquered surface of FEITENG panels is very flat and homogeneously smooth, providing outstanding ink adhesion. It is printable on all standard flatbed printers. All standard FEITENG colors, as well as most decorative FEITENG panel surfaces (e.g., FEITENG silver finish, FEITENG mirror, or FEITENG decor), are ideal printing substrates and have proven excellent printing results over the past decade.

FEITENG panels are resistant to temperatures from +80°C down to –50°C, as well as resistant to water and UV radiation. Yellowing or distortion caused by UV lamps does not occur. Therefore, digitally printed signs can be used long-term and maintain their color quality even outdoors.

Regarding pre-treatment, we recommend the same process as mentioned for screen printing. To avoid fingerprints on the surface, wearing gloves is the easiest solution. Various processing techniques such as routing, folding, or bending are possible with FEITENG panels – even after printing with solvent-based inks.

FEITENG

Screen Printing

Stove-lacquered FEITENG panel surfaces are well suited for screen printing. It is important to remove the protective film and to clean the surface using ethyl alcohol or isopropyl alcohol with a lint-free cloth prior to printing. The alcohol must not be applied directly to the panel. Between cleaning and printing, the alcohol needs approximately 10–15 minutes to evaporate. Lamination of the prints can be useful in order to achieve special surface effects or to improve mechanical or chemical properties.

New idea with new dimensions

The creative potential of FEITENG becomes obvious when processing the material. Small sizes, large formats, and even 2000 mm widths – FEITENG offers many possibilities. Even after printing with adapted inks, each format can be cut to size, shaped, formed, bent, or folded.

Especially for flat applications, FEITENG digital printing panels deliver high-quality imaging and exceptional efficiency. FEITENG digital features an optimised lacquer system for direct-to-substrate digital printing, providing excellent ink adhesion that allows advanced printing speeds. Higher performance means higher output and lower costs!

Applicable for flat applications such as

- Indoor and outdoor signage
- Hoardings
- Photo mounting and laminating
- Screen printing





MAINTENANCE & CLEANING

Touch-up Coating Method

Commercial or custom acrylic paints are suitable for repair coating of all finishes of FEITENG panels. Typical procedures are as follows:

- Clean the surface and remove any dirt or contamination.
- Stir the paint well. Apply the paint using a brush or a pencil-type touch-up container.
- Dry and cure at room temperature, following the manufacturer's instructions.

Normally, acrylic paints show good adhesion after curing. However, the touched-up portion may appear slightly different from the surrounding area. Since the final appearance depends on the coating method, even an exactly matched paint may show a slight difference to some extent.

For stone and timber finishes, use an intermediate solid color diluted with clear paint for touch-up. The suitable dilution rate depends on the color, typically 10–90% clear content. The Munsell number attached to each color can serve as a guide for selecting the appropriate intermediate color.

Precautions for Cleaning FEITENG

The surface of FEITENG composite panels requires regular, professional cleaning and maintenance. Such cleaning is not only to maintain a clean and attractive appearance, but also to periodically remove harmful contaminants that accumulate on the surface (such as industrial deposits, chemical pollutants, grease, salt, etc.), thereby preventing these pollutants from causing corrosion, accelerated aging, or permanent damage to the coating or other surface treatment layers over time. This ensures the expected decorative effect and protective function of the panels. FEITENG has detailed cleaning standards for various types of materials. For more information, please visit www.feitengacp.com to refer to the Precautions for Cleaning FEITENG Metal Composite Panels.

